

YouCred: An Online Tool Co-Designed with Fact-Checkers for Misinformation Discovery on YouTube

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Monitoring online platforms, one of the most challenging aspects of the fact-checking process, remains predominantly manual. Crafting search queries to surface potentially dubious online content is based on guesswork, demanding significant time and effort from fact-checkers. While monitoring tools exist for platforms like Facebook and X, a void exists for video platforms like YouTube. In this paper, we present YouCred, a system developed through a two-year partnership with Pesacheck, Africa's largest indigenous fact-checking organization. YouCred supports misinformation discovery by generating search queries for topics of interest to fact-checkers and offers an interface for flagging and annotating videos containing potentially misinformative claims. It addresses the real-world needs of fact-checkers and harnesses their unique assets, including indigenous knowledge, resources, and expertise. Through a nine-month field deployment, we examined how fact-checkers integrated YouCred into their workflows. We find that YouCred provided meaningful query suggestions, enabled more targeted video discovery, and increased the efficiency of their fact-checking workflow. We reflect on YouCred as a long-term, real-world case of collaborative system development and deployment—highlighting design implications, showing how sustained use revealed evolving needs, and underscoring the importance of continuous adaptation and maintainability in socio-technical systems.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing systems and tools**; *Human computer interaction (HCI)*; • **Information systems** → *Information retrieval*.

Additional Key Words and Phrases: fact-checking, misinformation, YouTube, participatory methods

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1 Introduction

Misinformation poses a critical challenge to societies worldwide, and YouTube plays a central role in its circulation [99]. Despite being the world's second-largest search engine [25] and often described by fact-checking organizations as a hub for conspiracy theories [99], YouTube provides few affordances to support fact-checkers' work [59]. The platform offers no features to track trending misinformative content, and identifying claims within videos remains a labor-intensive process [26, 59]. Despite repeated calls from fact-checking organizations for dedicated tools to support fact-checking on video platforms [26, 59], no widely adopted system currently exists. In

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this study, we address this gap by developing YouCred, a system designed to support fact-checking on YouTube.

In the past, academic research on fact-checking has had limited impact on the practices and reporting methods of fact-checking organizations [35, 59]. This is partly because the design of fact-checking systems is often framed as a primarily technical challenge. However, fact-checking is inherently a socio-technical phenomenon. It relies not only on computational methods to surface and analyze content, but also on the practices, judgment, and expertise of fact-checkers themselves [59, 98]. Sustainable socio-technical systems emerge when technical advances are paired with attention to the practices, relationships, and capacities of the communities who use them. Prior work highlights the importance of giving voice to fact-checking organizations in the design process to ensure that new systems cater to their specific real-world needs [59]. However, focusing just on needs, while essential, can inadvertently overlook the latent potential residing within an organization's assets—their expertise, knowledge, and resources [100, 102]. Recognizing and incorporating these assets can lead to design solutions that are not just functional but also culturally relevant, contextually attuned, and poised for sustainable impact [100, 102]. Thus, in this work, we partnered with Pesacheck, Africa's largest indigenous fact-checking organization, and adopted both need-based and asset-based participatory design approaches to develop YouCred. We engaged actively with fact-checking stakeholders at Pesacheck throughout the project to ensure that the system addressed their requirements while tapping into the organization's internal strengths, expertise, and contextual knowledge.

First, we conducted a formative interview study with eleven individuals from Pesacheck performing various fact-checking roles (as identified in [59]). These interviews aimed to gain insights into the current fact-checking processes, tools, and resources employed by the organization for monitoring YouTube, as well as to identify the platform-specific challenges. Consistent with prior research [59], our interviews revealed that fact-checkers rely heavily on manual searches to find misleading content on YouTube. In addition, the process of generating search queries that could lead to potentially dubious content is ad-hoc and based on guesswork, domain knowledge, and experience. As a result, fact-checkers spend a lot of effort coming up with search queries to monitor the platform. We also identified a lack of a video annotation tool to support video-based credibility assessments; fact-checkers instead maintained records using external tools such as Google Docs. Based on these insights, we designed the YouCred system to aid fact-checkers with misinformation discovery and video-level annotation on YouTube. The system is a result of a 2-year-long collaboration with Pesacheck. We integrated the knowledge and feedback of Pesacheck's fact-checkers throughout all stages of system development, including requirement elicitation, system design, deployment, and testing. YouCred includes two primary components: (1) a misinformation discovery module that generates search queries related to significant events and topics of interest to fact-checkers, and (2) an annotation interface to support structured, in-system video annotation aligned with fact-checkers' existing workflows.

To test the acceptance of YouCred, we deployed the system at Pesacheck for nine months. During this period, we closely monitored its usage and conducted interviews with Pesacheck's team. Our findings suggest that fact-checkers regularly used YouCred to monitor topics ranging from Ebola to election-related misinformation. The automatic query generation feature was especially valuable in surfacing relevant yet unexpected search terms, such as conspiracy-linked keywords, that fact-checkers might not have identified on their own. Additionally, the structured annotation interface replaced fragmented tools like Google Docs, enabling more systematic labeling of claims. At the same time, real-world deployment revealed evolving needs, underscoring the importance of continuous adaptation and maintenance in sustaining socio-technical systems. Overall, our study makes the following key contributions:

- Our work emphasizes the significance of an ongoing dialogue between fact-checking organizations and technology developers. We demonstrate that by integrating stakeholders' insights and leveraging their unique assets in the design of fact-checking systems, we can develop solutions that have a tangible impact on real-world fact-checking practices.
- Fact-checking research is often focused on the Western world, leaving a gap in research efforts in the Global South. YouCred fills the void through its collaboration with South Africa's indigenous fact-checking organization and by building a fact-checking system catering to the organization's unique challenges of finding misinformation on the YouTube platform.
- YouCred introduces a structured approach to search query generation by automatically suggesting terms related to emerging events and topics of interest. This approach reduces reliance on guesswork and supports a more systematic, scalable process for identifying potentially misinformative content on YouTube.
- We demonstrate the practical value and utility of YouCred for fact-checkers through an extensive 9-month deployment and evaluation. Our study also underscores that designing and deploying systems is not enough; continuous maintenance and evolution are essential to effectively address the evolving needs of fact-checkers.

2 Related work

2.1 The landscape of research in designing fact-checking systems

In recent decades, researchers have proposed a range of systems to automate and scale various aspects of the fact-checking process (see [75] and [108] for a detailed review). Since finding misinformative content remains one of the most challenging steps in workflow [59], many of these systems focus on monitoring online platforms [42, 70]. For example, *WhatsApp monitor tool* displays frequently shared multimodal content from public WhatsApp groups [70], while *Watch 'n' Check* monitors trending topics on Twitter and facilitates the discovery of tweets containing specific keywords [14]. In addition to the externally available tools, some platforms offer internal misinformation monitoring tools only to partner fact-checkers. Meta's *Queue*, for instance, surfaces user-submitted and AI-surfaced potentially misleading content on Facebook and Instagram [26]¹. In contrast, YouTube lacks both internal and external tools to help fact-checking organizations monitor the platform effectively. While YouTube does allow users to flag content and employs automated moderation systems—including machine learning models [105, 106], these mechanisms are designed for general content moderation and not for professional fact-checking workflows. Instead, YouTube relies on third-party fact-checking organizations to independently surface and verify misinformative content. These organizations use the ClaimReview schema to tag fact-checks they publish, which YouTube may then display in search results [19]. However, identifying such content on the platform remains a largely manual and unsupported task. Our work addresses this gap by designing a system that helps fact-checkers discover, annotate, and assess potentially misinformative videos on YouTube.

In addition to platform monitoring tools, previous research has explored various methods of determining fact-check-worthy claims in text [41, 55], including crowdsourcing [3, 4, 11, 32, 43, 61, 69], artificial intelligence [13, 30, 107], natural language processing [31, 97], knowledge graphs [88], visual analytics [64], or a combination of human assessments and AI predictions [53, 76]. End-to-end fact-checking systems like ClaimBuster Scholars have also been developed to catch factual claims in live discourse on social media and match them with a repository of fact-checks to determine their veracity [40]. Despite the plethora of available systems, only a fraction have demonstrated real-world impact [34, 59], often due to limited adaptability across contexts and a lack of attention

¹As of Jan 2025, Meta said it would stop using third-party fact-checkers on Facebook and Instagram.

to the practical needs and constraints of fact-checking organizations [59]. For instance, tools dependent on structured public data are difficult to use in regions lacking accessible databases [35]. To address these limitations, fact-checking systems must be designed around the specific needs, knowledge, and resources of the organizations that use them. Our work embraces this approach by supporting YouTube monitoring in partnership with a fact-checking organization, grounding the system in both practical constraints and institutional strengths.

2.2 Misinformation and fact-checking in the Global South

Misinformation represents a global phenomenon, yet the majority of research on misinformation and fact-checking has been narrowly focused on cases in the English-speaking Western world, leaving a significant portion of this issue unaddressed in Global South countries [38]. Scholars attribute this oversight to various factors, including a prevailing bias in mis/disinformation research that disproportionately focuses on threats to the interests of white communities. This bias is rooted in historical and systemic structures shaping research priorities, reflecting a broader trend in academia where the perspectives of white populations often take precedence, potentially perpetuating existing power dynamics and overlooking the nuanced impact of disinformation on non-white communities [2, 78]. Another contributing factor to the under-representation of the Global South is the potential lack of cultural or linguistic knowledge among researchers required to conduct studies in these communities [86]. However, in recent years there has been a rising interest within the Human-Computer Interaction community to explore the phenomena of mis/disinformation in non-Western regions worldwide, including Vietnam [73], India [89, 96], Brazil [65, 84], Bangladesh [38, 93], Sub-Saharan African countries [95], and Nigeria [7], among others. Scholars in these studies often employ community-engaged methods to investigate how communities in the Global South interact with misinformation [96], explore the practices and challenges faced by professional fact-checking organizations in the Global South [8, 59], and increasingly adopt participatory design to empower marginalized communities in shaping the design of tools and systems for their use [86]. Our work contributes to this growing body of research. By employing participatory design methods, we actively give a voice to the fact-checking community at Pesacheck in shaping the design and development of the YouCred fact-checking system.

2.3 Design approaches for building socio-technical systems

Researchers in the field of Human-Computer Interaction (HCI) have been increasingly delving into the intricacies of design processes and methodologies for constructing socio-technical systems. Among the prominent approaches is user-centered system design (UCSD), which employs an iterative design approach, placing users' input and needs at the core of each design phase [77]. In UCSD, designers utilize a myriad of investigative and generative methods, including interviews, surveys, contextual inquiries, brainstorming, and more, to develop a deep understanding of users' needs and preferences [5]. In contrast to UCSD, users take a more active role in participatory design (PD). PD allows the inclusion of stakeholder groups, who are the intended users of the system, in the design process through discussions and prototyping [9, 36, 74]. PD introduces community-centered democratic values in the design process by valuing each participant's input in designing and refining the system, which cultivates a strong sense of ownership among end users [86]. Both the aforementioned design approaches take a "*needs-finding and needs-solving*" perspective, helping researchers to identify the needs or problems faced by the stakeholders and designing solutions to address those needs [46]. However, scholars posit that such a need-focused design approach can sometimes fall short to produce a long-term sustainable solution, particularly in resource-constrained contexts, potentially leading to dependence on external aid [10, 50, 82, 91, 100]. Furthermore, the needs-based lens may perpetuate a negative narrative by

focusing on the ‘*deficits*’ rather than the strengths and potential of the stakeholders. To counter these limitations, researchers advocate for a shift from a focus on ‘*needs*’ to ‘*assets*’ and proposed Asset-based PD as an alternative design approach [17, 49, 82, 100, 102]. The assets-based approach centers on harnessing the resources, knowledge, and capacities of stakeholder groups to formulate solutions. This approach has gained traction in designing socio-technical initiatives, particularly those aimed at marginalized populations [49, 100, 101, 104]. In this study, we understand the existing assets—knowledge, methods employed, and resources—held by the fact-checking community at Pesacheck to develop fact-checking solutions. We avoid an exclusive focus on either a needs or assets-based approach, opting to consider both. Our design remains driven by stakeholder needs, while the solution is constructed upon the assets owned by the fact-checking community.

3 Formative study

To inform our work, we conducted a formative interview study to understand Pesacheck’s current fact-checking practices to monitor YouTube and opportunities for improving those practices. In this section, we describe the details of our formative study including the participant details (Section 3.1), study procedure (Section 3.2), and findings (Section 3.3). Next, we describe the design goals we identified to guide the development of YouCred (Section 3.4). Finally, we describe the collaborative and iterative design process adopted for our study (Section 3.5). It is important to note that we conducted the formative interviews only after our study design was reviewed and approved by our university’s Institutional Review Board.

3.1 Participants and Procedures

We employed convenience sampling [24] to initiate recruitment for the formative study, drawing on the senior author’s established working relationship with senior staff members at Pesacheck. We then used snowball sampling within the organization, as these initial contacts introduced us to other individuals involved in fact-checking at Pesacheck. In total, we conducted interviews with 11 individuals (P1-P11) who held various roles within the organization. These roles included eight fact-checkers responsible for monitoring online platforms and investigating dubious claims; two news editors overseeing the fact-checking process, including decisions about which topics to prioritize; and one investigative researcher focused on tracking and analyzing persistently circulating misinformation. For a detailed understanding of these roles, please refer to [59]. To preserve our participants’ anonymity, we refrain from providing their demographic details and designations within the organization.

3.2 Interview protocol

We began the interviews by asking participants how they monitor YouTube and discover misleading videos on the platform. Then, we asked participants to share their work screens and guide us through a fact-checking report debunking a false claim made in a YouTube video. We asked them to explain the entire process followed in the report, including how they found the video, identified the problematic claim, conducted investigative work, and wrote the fact-checking report. Next, we asked questions about the tools that fact-checkers employed in the process and their limitations. Finally, we asked about the challenges encountered throughout the fact-checking process on YouTube and discussed how the affordances of YouTube facilitated or hindered their work. Overall, the formative interview study provided valuable insights into the current processes, needs, and challenges experienced by the stakeholder groups. We conducted interviews on Zoom and they lasted between 60-90 minutes. The first and last authors independently reviewed the interview transcripts and conducted a thematic analysis using an inductive, iterative approach. Any discrepancies in coding were discussed and resolved collaboratively.

3.3 Findings

In this section, we provide an overview of the key themes that emerged from our formative study.

3.3.1 YouTube—a blindspot for Pesacheck. YouTube has been a *blindspot* for Pesacheck organization for a very long time. Participants reveal that they are aware that a lot of misinformation circulates on the platform but “*they have seen a decline in the YouTube videos that [they] fact-check because of the hardships that [they] face with finding misinformation on the platform*” (P5).

“YouTube is one of our blind spots. YouTube has been used for disinformation for quite a long time right now. But we don’t have access or visibility to YouTube unless you go and manually use the [YouTube] search button. -P2 ”

3.3.2 Searching for misinformation is manual, cumbersome, and based on guesswork. The interviews revealed that searching for misleading content on YouTube remains a manual and difficult task. The generation of search queries to find potentially misleading content is based on “*guesswork and trial-error method*” (P3). As one participant explained, “*To search about a particular topic/claim on YouTube, fact-checkers start with relevant terms related to certain claims or terms that best describe that person or their situation or history, and [they] keep trying different options until [they] find some results that they’re looking for*” (P4). P3 further added that searching for generic queries like “*Obama will get both misinformation and non-misinformation content. So it’s up to [the fact-checker] to go through all search results to figure out which one contains misinformation*”. Going through the search results is also challenging because the top search results usually contain videos from mainstream news channels. As one participant elaborated, “*If you look for, say, Ebola, they’ll give you these videos in the top that are authentic, say, from the BBC, Aljazeera, and CNN, and that is not potential misinformation*” (P5). As a result, fact-checkers mostly discover and fact-check videos when they find a YouTube link embedded in a Facebook or Twitter post or when users report it via their tip lines². Then they go to YouTube to find similar videos. As P1 explains, “*For us to find the misinformation on Youtube, we would go on other social media platforms such as Facebook, Twitter. And once we identify a video that has some misinformation, we go on Youtube and find. you know, a similar video*”.

3.3.3 Challenges of video modality. Participants (P1-4, P7-8) revealed that the modality of videos poses additional challenges, as fact-checkers may have to “*watch very long videos from start to end*” (P2). To save time, some fact-checkers prefer using textual transcripts and searching for specific keywords. As one participant explained, “*We go to a video that is 1 hour long and if you’re looking for misinformation about COVID-19, we just search for COVID-19, and then we only go through the instance where the person in the video spoke about COVID-19*” (P3).

3.3.4 Limitations of YouTube’s affordances. Interviews revealed that the YouTube platform also lacks certain affordances that are useful for fact-checking workflows, for example, “*the ability to search for a lot of things at once*” (P2), “*get the analytics for search results*” (P2), ability to download search results (P2, P3), and the capability to filter search results using a combination of search filters, for example, date-range and engagement metrics (P2-5, P7-9).

3.3.5 Unstructured Video Annotation Process. The interviews further revealed that the process of annotating a video for veracity is unstructured and they rely on Google Docs to keep track of all information about the fact-check (P1-3, P7). P1 explains—“*When it comes to video annotation, it is*

²Online platform users can report instances of misinformation to Pesacheck through various channels. The organization has established a WhatsApp number for user submissions, and alternatively, reports can be made via email or through Pesacheck’s official accounts on popular social media platforms such as Twitter, Facebook, and Instagram.

very analog. What we do is take the actual video, download the transcript..figure out what part of the video we want and use google docs to annotate the videos”.

3.3.6 Desired Features in the fact-checking system for YouTube. When probed about what kind of tool they envision, we repeatedly observed participants expressing the need to have control over the outcomes of the tool, for example, *“I want to be able to choose the keywords”, “I need the ability to decide what videos I want to look into”, and “humans need to come in to verify, I don’t think any tool can really do exactly what we can, not yet”*. All of these statements underscored participants’ strong desire for agency over the tools they would utilize. Participants also expressed a desire for the tool to support searching *“in a specific country, a specific region”* (P5) and the ability to search in *“regional languages of Africa, such as Amharic”* (P1-5, P7-9). They also desired a unified system that incorporates most elements of their workflow, eliminating the need to switch tabs and providing a seamless experience (P4, P6-8).

3.3.7 Building upon fact-checkers assets—knowledge, practices, and resources. All participants emphasized the importance of a system that seamlessly complements rather than replaces their existing practices. During our interviews, we explored the search strategies participants employ on other social media platforms, such as X, that have proven effective. They highlighted the efficacy of searching for trending content related to a topic, considering it representative of people’s current interests. Furthermore, they also described using their deep local knowledge about events, narratives, and regionally specific claims to surface relevant content. These practices and strategies directly shaped how we approached search query generation for the YouTube platform.

Participants also revealed their practice of maintaining an internal database of previously discovered misleading videos, often embedded in posts on other platforms. These fact-checked videos are diligently archived, ensuring their availability even if removed from the platform. Despite recognizing the potential richness of this data collection, participants acknowledged its underutilization, underscoring the untapped resources that YouCred could effectively leverage.

Interviews also revealed that Pesacheck’s fact-checking team possesses strong investigative expertise, including identifying misleading claims and corroborating them through expert consultation, in-depth research, and the use of specialized tools. They also demonstrated a deep understanding of local contexts, cultural nuances, and regional issues that are crucial to assessing the accuracy of claims within specific geographic settings. Their multilingual proficiency emerged as a valuable asset, enabling them to engage with content in local regional languages effectively. Rather than replacing these skills, YouCred is designed to support and amplify them by helping surface potentially misinformative content in specific languages and regions of interest and by offering structured video metadata and transcripts to aid in the evaluation of video content.

In Section 3.4, we elaborate on the design goals of YouCred, shaped by the identified needs and preferences of fact-checkers. Subsequently, in Sections 4, 5, and 6 we detail the various features of YouCred, developed under the guidance of fact-checkers, leveraging their assets.

3.4 Design goals

Drawing from previous research [59] and our formative study, we identified six design objectives that inform the design of YouCred. These objectives directly address the needs and obstacles encountered by the fact-checking team at Pesacheck and are built upon their existing strengths and assets. (Table 1).

- *Automate search query generation:* YouCred aims to automate the generation of search queries, which is currently a tedious and manual task for fact-checkers. The system eliminates guesswork by suggesting relevant keywords that could lead to misinformative search results.

Challenges & Needs	Design Goals	Corresponding Asset
Searching for misinformation is manual, cumbersome, and based on guesswork	Automate search query generation	Established search strategies and curated database of misleading videos
Watching long videos for credibility assessment is time-consuming, and video annotation process is unstructured	Design an annotation interface that simplifies and structures the credibility assessment process	Proficiency in investigative work and annotation practices
No ability to search for multiple queries simultaneously	Provide multi-term tracking	Ability to monitor multiple streams of information, including various narratives and topics
Absence of comprehensive analytics for search results	Display engagement patterns and upload dates via intuitive data visualizations	Ability to identify patterns and prioritize content
Limited options for filtering search results using multiple criteria like date-range and engagement metrics	Enable simultaneous filtering of search results using various criteria, including engagement metrics and publication dates	Multitasking and search optimization practices
Need for the system to support local regional languages	Include support for local and regional languages	Multilingual capabilities and regional knowledge
Need for agency in the system	Strike a balance between automation and stakeholder expertise	Expert judgment and contextual understanding

Table 1. Alignment between stakeholders' challenges and needs, design goals for the YouCred system, and corresponding fact-checker assets.

These keywords are generated through four distinct methods, each designed to align with fact-checkers' established search practices and strategies and to leverage a curated database of previously identified potentially misleading videos (Section 5.2).

- *Provide structure to credibility assessment process:* Fact-checkers are skilled at examining content, identifying misinformative claims, and conducting investigative work. YouCred facilitates these tasks by offering a structured annotation interface, streamlining their ability to analyze videos and transcripts and efficiently curate and organize information, enhancing the overall credibility assessment process.
- *Data visualization to provide insights and facilitate prioritization:* YouCred empowers fact-checkers with intuitive and informative data visualizations that present crucial engagement patterns (such as likes, views, and comment counts) and publication dates in a clear and accessible manner. This feature enables fact-checkers to efficiently prioritize and filter videos for analysis.
- *Multi-term tracking:* Fact-checkers expressed a desire to track multiple search terms simultaneously. By providing this functionality, YouCred aims to facilitate efficient monitoring and analysis of content relevant to their topics of interest.
- *Adaptability to different languages and contexts:* YouCred addresses the fact-checkers' desire for multilingual content discovery and annotation capabilities by supporting search in various

languages and in specific regions. This feature is grounded in fact-checkers' deep knowledge of regional contexts and languages, enabling more localized misinformation tracking.

- *Agency and human involvement:* Fact-checkers value their agency in the fact-checking process and consider human involvement crucial for verification and contextual understanding. YouCred aims to strike a balance between automation and human expertise, allowing fact-checkers to have control over verification capabilities while leveraging the benefits of automated tools.

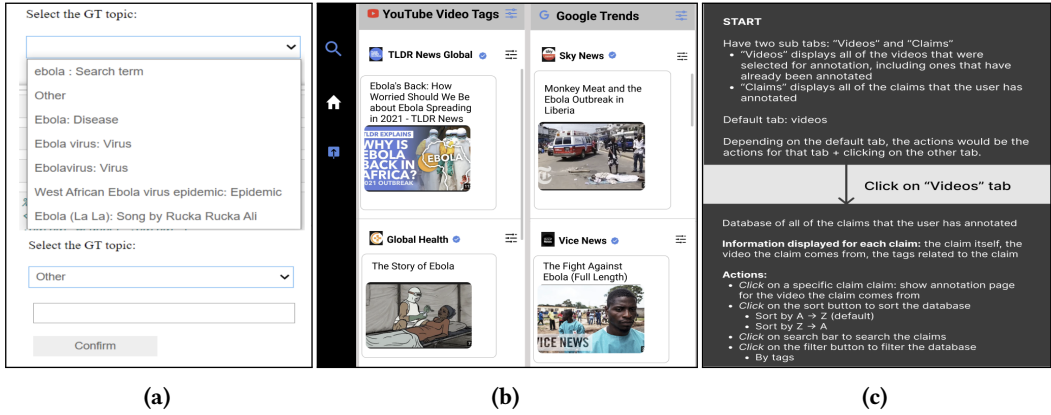


Fig. 1. (a) A snapshot of the UI widgets implemented in the Jupyter Notebook to demonstrate the search query generation methods, (b) Initial wireframe of the YouCred *view-results* page developed in Figma, (c) Snapshot of an early workflow diagram created for YouCred.

3.5 Design process

We adopted a collaborative and iterative design approach to develop YouCred. To explore possible directions, the first author implemented early versions of the search query generation methods, drawn from prior literature [59] and formative study findings, as interactive widgets in a Python Jupyter Notebook (e.g., Figure 1a). This notebook served as a hands-on demo tool to present ideas and gather early feedback from Pesacheck's fact-checking team, who expressed strong interest in supporting the system's design and evaluation.

Working alongside two experienced UX designers, we then created wireframes (e.g., Figure 1b) and workflow diagrams (e.g., Figure 1c) using the widely used prototyping tool, Figma³. The development of YouCred then commenced, with features being added and refined iteratively. Throughout the project, the Pesacheck team generously shared their time, expertise, and ideas, demonstrating continuous engagement. We conducted approximately 38 Zoom meetings (each lasting 1–2.5 hours, with 1–21 participants) and maintained day-to-day communication via Slack. Appendix Figure A.1 presents an overview of the project timeline. In the next section, we describe the resulting YouCred system in detail.

4 Overview of YouCred

YouCred is a fact-checking system specifically developed to support fact-checkers in combating online misinformation on the YouTube platform (see Figure 2 for an overview). At its core, YouCred

³<https://www.figma.com/>

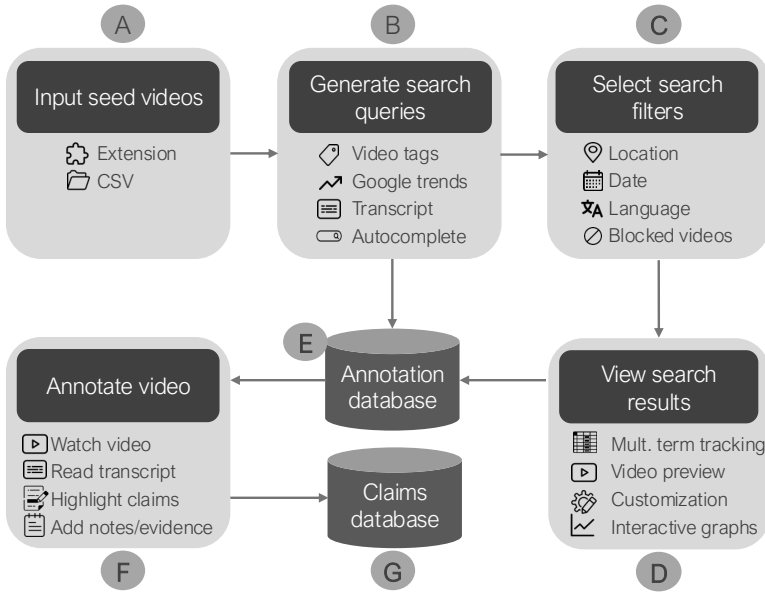


Fig. 2. Figure illustrating YouCred’s workflow. (A) To begin, fact-checkers are required to input seed videos related to the topic of interest via extension or by uploading a CSV. (B) Fact-checkers then select one or more query generation methods and (C) can fine-tune their searches by specifying search filters. (D) The search results corresponding to all the generated queries can be viewed on the *view-results* page. On this page, fact-checkers can filter videos using interactive graphs, edit search queries, add additional queries to track, and preview videos. (E) Potentially misleading videos discovered in search results can be cataloged in the annotation database. Fact-checkers can later revisit this database to select videos for fact-checking. (F) Each video in the database leads them to its annotation page, where they can highlight potentially misleading claims and assess their credibility. (G) All highlighted claims, along with their veracity labels and other important metadata, are systematically stored in the claims database for future reference. At present, YouCred does not support multiple user accounts. While the tool is designed for individual use, it utilizes a collaborative database, enabling each user to access queries searched and videos annotated by other users.

supports content discovery for fact-checking by generating search queries aligned with fact-checkers’ monitoring goals and surfacing candidate videos for investigation on YouTube. By automating search query generation, YouCred reduces reliance on ad hoc, manual search methods and supports more structured content discovery. Additionally, it includes a video annotation interface that allows fact-checkers to preview video transcripts, highlight potentially misinformative claims, and store supporting notes and evidence. In the following sections, we provide a detailed description of YouCred, focusing on how it facilitates the discovery and investigation of potential misinformation (Section 5) and video annotations for credibility assessments (Section 6).

5 Misinformation discovery

The primary objective of the YouCred system is to help fact-checkers discover potential misinformation on the YouTube platform. The system accomplishes this objective by generating search queries via four methods, including leveraging YouTube video tags, Google Trends search queries, YouTube’s autocomplete suggestions, and distinctive terms from the transcripts of misinformative videos. Each method offers a different lens for surfacing relevant search terms—whether by capturing the language used by content creators or reflecting the queries commonly entered by

users seeking information on the topic. To initiate the process, fact-checkers must provide a small, curated list of potentially misinformative *seed* videos related to a particular topic as input. These videos serve as contextual anchors, helping the system extract topic-relevant terminology and generate effective search queries. While only a minimum of one seed video is required for a topic, we recommend that fact-checkers upload at least three videos for each topic to get better results based on our testing. We describe how fact-checkers input videos into the system in Section 5.1, and detail the four query generation methods in Section 5.2.

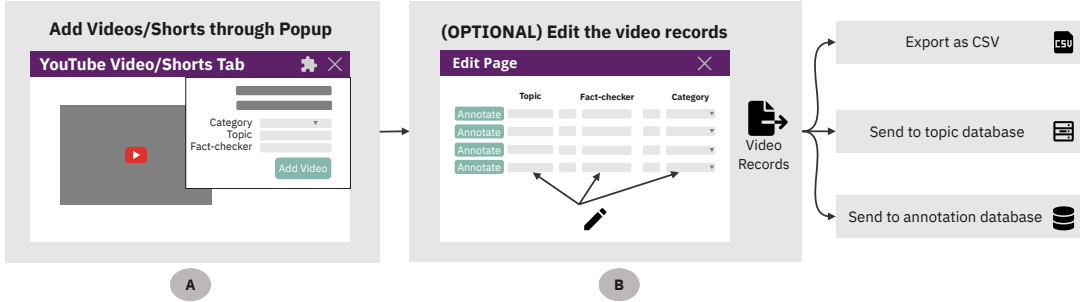


Fig. 3. Figure illustrating the workflow of YouTube-CSV-Helper extension. (A) When fact-checkers encounter potentially misleading YouTube videos, they can click on the extension that triggers a pop-up displaying the video's title and URL. Within the popup, fact-checkers can assign a category indicating veracity label (default is 'No conclusion'), select or create a topic, and enter their name for attribution. (B) The extension also provides access to an 'Edit' interface where fact-checkers can manage all their added videos. From here, they can export data as CSVs, send videos to YouCred's topic database (Appendix Figure A.2) for monitoring, or the annotation database (Section 6.1) for credibility assessment.

5.1 Inputting seed videos to YouCred

Fact-checkers can input seed videos into the YouCred system in two ways: by manually uploading a CSV file or utilizing the YouTube-CSV-Helper (YCH) Chrome extension. Initially, only the CSV option was available. However, in response to feedback about the tediousness of creating CSVs for each topic, we developed the YCH extension to simplify curating and inputting seed videos. Figure 3 describes the extension workflow. Once uploaded, using either method, the seed videos are stored in a central topic database (Appendix Figure A.2) and remain accessible for future use.

5.2 Formation of search queries

YouCred offers four distinct methods for query generation, each designed to align with fact-checkers' established practices. Importantly, fact-checkers retain full agency over what is searched: they can freely add, modify, or remove suggested search terms before initiating a search on YouCred. To refine results, fact-checkers can specify optional parameters such as region and language, which narrow search results to specific countries or languages⁴. They can also set the date range to retrieve videos published within a specific time frame. Additional customization includes setting the number of desired results and selecting search filters such as upload date, view count, relevance, or video rating such as upload date, view count, relevance, or video rating. These features (Appendix Figure A.3) allow fact-checkers to tailor search queries to their specific monitoring needs. The four query generation methods are detailed in the following subsections.

⁴These filters are supported via the YouTube API used by YouCred: <https://developers.google.com/youtube/v3/docs/search/list>. They are not available when manually searching on the YouTube platform.

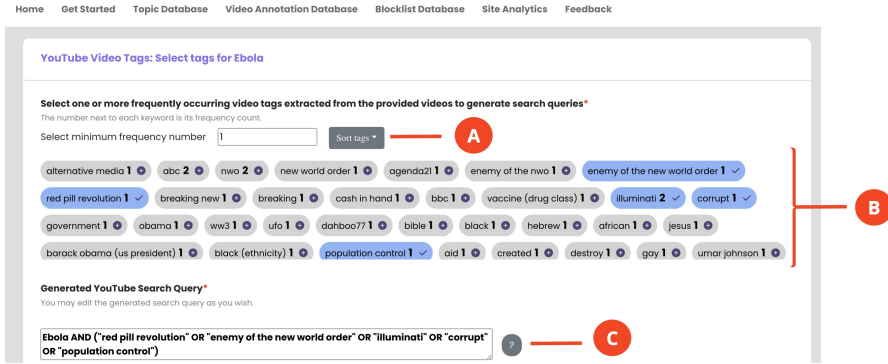


Fig. 4. Figure illustrates YouCred’s query generation method page, which utilizes the YouTube video tags method. The page displays a collection of video tags present in the seed videos for the topic *Ebola*, such as ‘enemy of the new world order’, ‘red pill revolution’, and ‘population control’. (A) The tags shown can be sorted either by frequency or alphabetically. (B) Each tag is accompanied by its frequency of occurrence. When a tag is selected, its corresponding bubble changes color to blue. (C) Fact-checkers can choose multiple tags, and as they make their selections, the chosen tags are appended with the topic to form the search query. Importantly, the search query is editable, allowing fact-checkers agency to modify it as needed.

5.2.1 YouTube video tags. We first utilize YouTube video tags found within the seed videos uploaded to YouCred as search queries. Tags are added by content creators to YouTube video’s metadata and are accessible via YouTube API⁵. Generally, tags consist of keywords or phrases intended to help YouTube’s search and recommendation algorithms understand the content of the video and enhance its discoverability [80]. Prior research has highlighted the significance of tags, both as input features for identifying potentially misinformative videos [79] and as effective search queries for retrieving potentially misinformative results on YouTube [57]. Thus, we use tags to assist fact-checkers in understanding the misinformative narratives associated with a particular topic and discover additional misinformative videos on the platform. Figure 4 illustrates YouCred’s query generation interface using the tag method.

5.2.2 Google Trends. Fact-checkers actively monitor online platforms by tracking current events and real-time search trends of people [59]. During the design process, fact-checkers expressed their interest in monitoring YouTube using popular search queries. To facilitate this, we leverage the Google Trends (GT) platform that provides real-time search trends on Google-owned platforms, including YouTube. The search queries obtained from GT have been previously used to audit online platforms for misinformation [16, 48, 57, 58, 83, 85]. YouCred also utilizes GT’s search queries to help fact-checkers gain a better understanding of the public’s interest and the prominence of certain claims on the YouTube platform (Appendix Figure A.4). YouCred extracts and displays both the most popular and least popular search queries related to a topic in the selected region and time period on the YouTube platform. This includes the commonly used terms by users as well as the less popular terms, which conspiracy theorists might exploit to spread false information—a phenomenon known as data voids [33].

5.2.3 YouTube transcript. Misinformative videos often harbor numerous false claims, which might not be evident in video tags. To address this, YouCred utilizes the keywords occurring in the misinformative seed videos as potential search queries to find other potentially misinformative

⁵<https://developers.google.com/youtube/v3/docs/videos>

The screenshot shows the 'YouTube Autocomplete Suggestion for: kenya elections' page. It includes sections for selecting top/bottom keywords, adding custom keywords, selecting suggestions, and generating a final search query. Red letters A, B, C, and D are overlaid on the interface to highlight specific steps: (A) points to the keyword selection area, (B) points to the custom keyword input, (C) points to the suggestion selection area, and (D) points to the final generated query.

YouTube Autocomplete Suggestion for: kenya elections

Select one or more frequently occurring keywords extracted from the provided videos to generate search queries*

Select top % of keywords: Sort tags ▾

kenya ruto raila secret uhuru chebukati copyright fair fair use iebc kenyacitizntv kumekucha leader mutembeiv

Select bottom % of keywords: Sort tags ▾

use tips version version raila video sponsored video version votes votes mt vs vs dynasty wa wa kabando wafula wafula chebukati

(Optional) Add additional custom keywords (separated by comma)

kenyan elections Get Autocomplete Suggestions

Select one or more YouTube Autocomplete Suggestion

Please note: YouTube may not have any autocomplete suggestion for your keyword.

william ruto macron 1 william ruto raila odinga 1 ✓ kenyan elections 1

Generated YouTube Search Query*

You may edit the generated search query as you wish.

kenya elections AND ("william ruto raila odinga" OR "kenyan elections 2022" OR "william ruto speech")

Fig. 5. YouCred’s query generation method page utilizing the autocomplete suggestion method. (A) The first step involves identifying seed words and phrases for extracting autocomplete suggestions. YouCred offers fact-checkers a choice of the most and least frequent $x\%$ unigrams and bigrams derived from seed video titles and descriptions. By default, x is set to 5, though fact-checkers can freely adjust this value. (B) Custom seed words can also be added. (C) YouCred collects all search queries related to these keywords and allows fact-checkers to select the ones they wish to monitor, (D) combining them using the OR search operator.

videos related to the topic (Appendix Figure A.5). To facilitate this, YouCred extracts transcripts of all the seed videos using the Python open-source package, *youtube-transcript-api* and applies standard text preprocessing (stop word removal, lemmatization, etc.). Then, it utilizes TfIdf to extract the unigram and bigram features. The usage of TfIdf features in the analysis of misinformative YouTube videos has been widely adopted and recognized in literature [27, 44, 57, 87]. TfIdf assigns higher weights to terms that are frequent within a specific transcript but relatively rare across the entire collection. This helps distinguish significant terms that are indicative of the content and themes of the seed videos. The fact-checkers can choose from the top/bottom $x\%$ of the features (default value of x is 5% but can be modified). Selected features are combined with OR operators so YouTube returns videos matching any term.

5.2.4 YouTube autocomplete. Autocomplete suggestion is a widely used functionality found in web search engines, designed to aid users in constructing their search queries. It is estimated that about 75% of search queries on Google are influenced by autocomplete suggestions [1]. Prior studies have shown that auto-complete suggestions could lead users to rumors online [45, 48, 58]. Additionally, previous work has revealed that autocomplete suggestions in languages of the Global South, such as Amharic, Kiswahili, and Somali, could especially expose users to harmful content [18]. Given their impact, misinformation appearing in search results for these terms could impact a large number of people and thus, are of interest to fact-checkers. Therefore, YouCred also allows fact-checkers to track autocomplete search suggestions for topics of interest (Figure 5).

5.3 Viewing and filtering search results

Once search queries are generated using one or more query generation methods, fact-checkers can access and evaluate the corresponding search results on the *view-results* page (Figure 6). The page

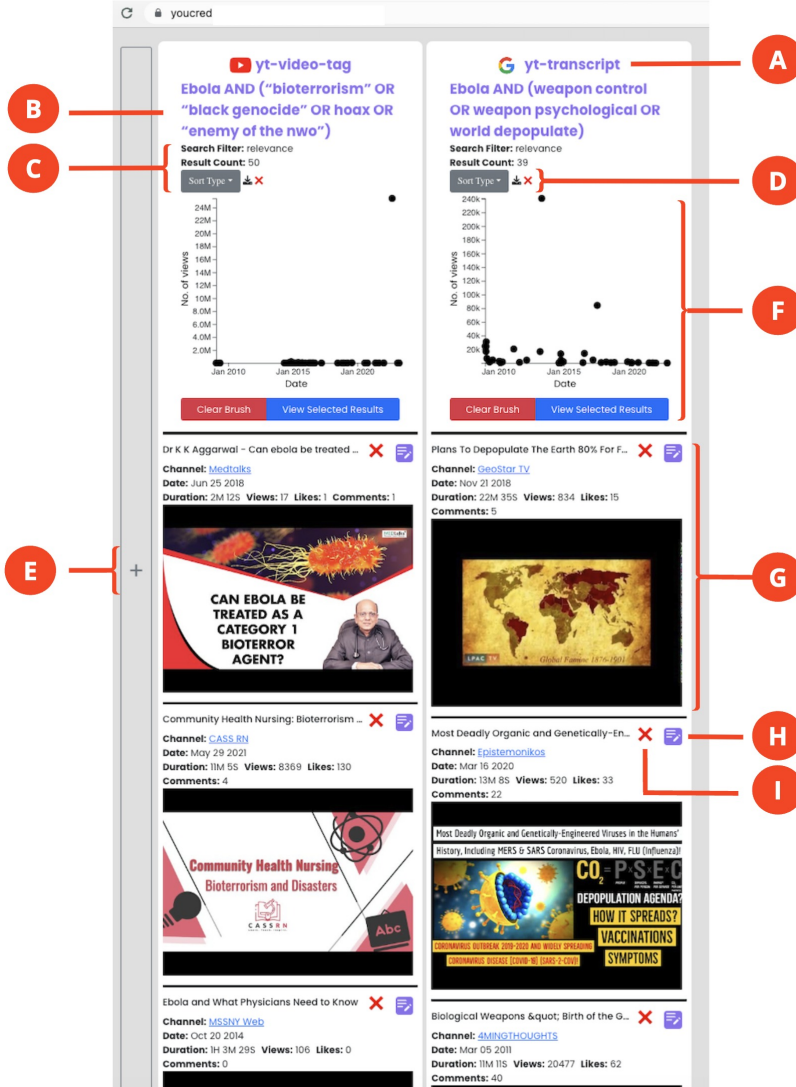


Fig. 6. Snapshot of YouCred’s *view-results* page that assists fact-checkers in tracking search results for multiple queries simultaneously. The page consists of multiple columns, with each column representing the search results of a specific query. The column header provides essential information, including the search query generation method (A), the search query itself (B), the applied sorting filter, and the count of search results (C). The page also offers functionalities such as downloading the search results as a CSV file and removing individual columns (D), or adding new columns (E) as needed. The interactive graph (F) within each column visualizes the engagement received by the search result videos and their corresponding publication dates. This dynamic graph allows fact-checkers to identify trends, and patterns, of engagement based on the age of the videos. Following the graph, the page includes sections dedicated to individual videos, representing each search result (G). These video sections provide important metadata, such as the video title, channel name, upload date, views, likes, comments, and a thumbnail. In cases where fact-checkers identify a potentially misinformative video, they can take action by adding it to the annotation database (H). This feature enables them to track and later fact-check the video. Additionally, if a fact-checker wishes to block a video to prevent it from appearing in future search results, they can utilize the block video functionality (I).

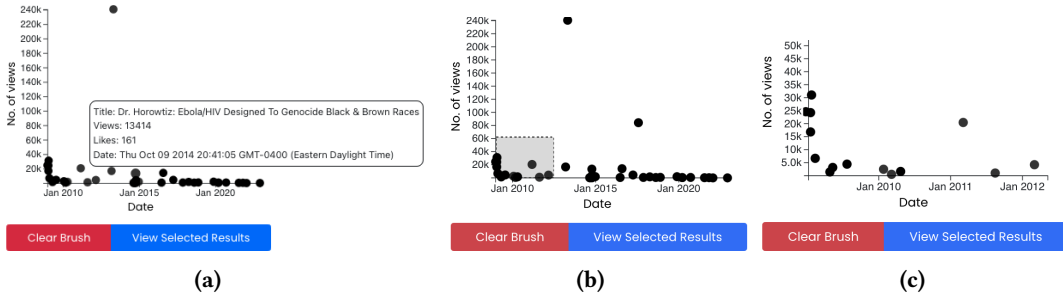


Fig. 7. The *view-results* page in YouCred features an interactive scatter plot graph that visualizes engagement data from the search results. The y-axis represents the number of views, while the x-axis shows the videos' publication dates. (a) Hovering over any point reveals a tooltip with the video's title, views, likes, and upload date. (b) Fact-checkers can "brush" (click and drag to select) a specific region of interest within the graph. (c) This action zooms into the selected area, offering a more focused analysis. The "View Selected Results" button filters the search results to include only the highlighted videos. To revert to the full set of search results, users can click the "Clear Brush" button.

offers a range of capabilities and features desired by fact-checkers to enhance their fact-checking workflow. We discuss them below.

5.3.1 Tracking multiple search queries simultaneously. The *view-results* page acts as a centralized hub allowing fact-checkers to monitor and manage multiple search queries simultaneously. This capability enables them to efficiently compare and evaluate various sets of search results, conduct comprehensive assessments of a topic, and respond promptly to emerging misinformation trends. Figure 6 shows the snapshot of the *view-results* page with two columns, each corresponding to a different search query. The header of each column denotes the query generation method, followed by the search query, the search filter selected to sort the results, and the number of search results. Fact-checkers can refresh and get the latest search results by simply double-clicking on the query and pressing the enter key.

5.3.2 Excluding Videos from Credible Channels. Fact-checkers observed that YouTube's top search results prominently feature videos from mainstream channels. In response, they compiled a list of 46 credible mainstream news channels whose videos have a lower likelihood of being misinformative, such as The Ethiopian Reporter, Nation Africa, New York Times, etc. Fact-checkers requested that videos from these channels be excluded from the displayed search results. YouCred achieves this by including a minus operator [72] before each channel name and appending it to the search query⁶. This modified query is then used to query YouTube and retrieve the search results.

5.3.3 Customization through editable queries, sorting options, and addition of new columns. YouCred offers fact-checkers the ability to edit their search queries with a simple double-click. Additionally, each column in the *view-results* page features a *Sort Type* drop-down button, allowing fact-checkers to further sort the fetched results on top of the previously selected filter based on criteria such as date, views, likes, or comments. YouCred also simplifies interface management by allowing addition of new columns to track additional search queries.

⁶For example, to exclude videos from specific channels, YouTube allows users to employ this minus operator, resulting in a query like: `search_query -channel_name1 -channel_name2 -channel_nameN`. The modified query is not visible to user.

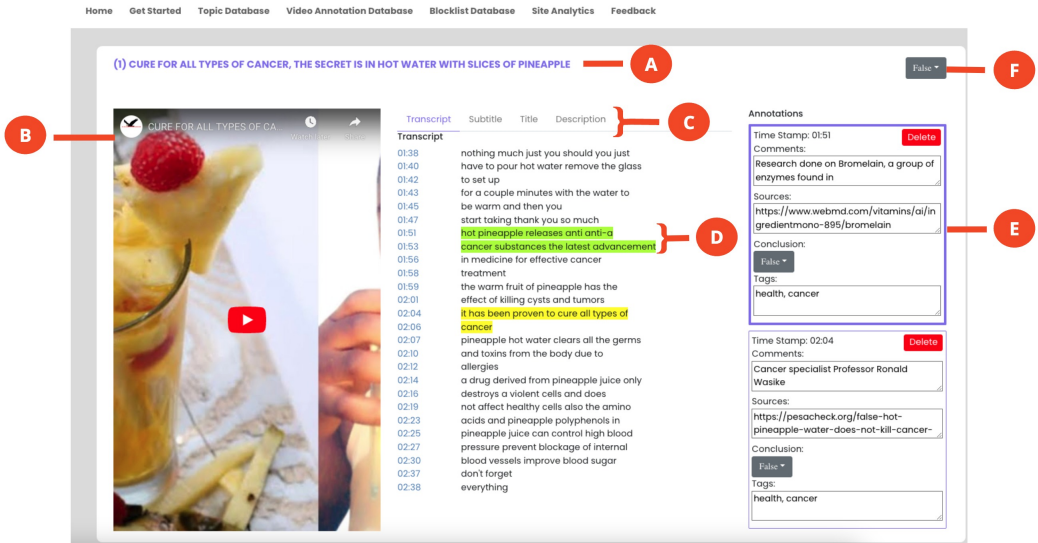


Fig. 8. YouCred’s annotation interface that streamlines and facilitates the credibility assessment process. (A) The header corresponds to the video’s title. (B) The video is embedded on the left side of the interface. (C) The video’s textual metadata, transcript, subtitles, title, and description appear in separate tabs in the center. (D) Fact-checkers can highlight potentially misinformative claims within any tab, (E) add annotations including comments, sources, and tags, and (F) assign an overall veracity label to the video.

5.3.4 Interactive graphs for engagement metrics. In response to fact-checkers request for visualizing search result analytics and filtering videos based on both engagement and date, each column in YouCred’s *view-results* page features a dynamic, multifunctional, and interactive scatter plot graph (Figure 7). The graph plots engagement metrics against video publication dates, with view count shown by default and options to switch to likes or comments. Each point represents a video, with hover interactions revealing key metadata. Fact-checkers can identify patterns and high-engagement clusters, zoom into regions of interest via brushing, filter to selected results, and reset the view as needed.

5.3.5 Video metadata sections. Below the interactive graph in each column, fact-checkers have access to detailed video sections containing search result videos along with their corresponding metadata, including title, channel name, upload date, views, likes, comments, and a thumbnail image. By clicking on the video thumbnail, fact-checkers can enter preview mode and view the video within the YouCred system (Appendix Figure A.6). If a video seems potentially misleading based on its metadata and preview, fact-checkers can add it to the annotation database (see Section 6.1) for investigation by clicking on . They also have the option to block non-misinformative videos from appearing in future searches by clicking on . The blocking feature streamlines their efforts by focusing on potentially misinformative videos, optimizing time and resources.

6 Misinformation Annotation and Evaluation

YouCred assists fact-checkers in systematically evaluating videos and labeling potentially misinformative content through three key components. First, it offers an annotation database (Section 6.1) that stores videos flagged for review. Second, it provides an annotation interface (Section 6.2) where fact-checkers can assign veracity labels and take notes. Third, YouCred curates a claims database

(Section 6.3) that consolidates annotated claims extracted from reviewed videos. We discuss these components in detail below.

6.1 Video annotation database

YouCred includes a dedicated annotation database (Annotation Figure A.7) to support fact-checkers in systematically storing videos they intend to evaluate at their convenience. Videos can be added to this database from the *view-results* page or the YCH browser extension. Each entry includes an ‘annotate’ button, which directs users to the corresponding annotation interface for detailed credibility assessments (Section 6.2). The database stores key metadata such as video title, engagement metrics, and channel information. The ‘conclusion’ column remains blank until a veracity label is assigned via the annotation interface. Additionally, an ‘added date’ column records when each video was incorporated, helping fact-checkers organize and track their entries over time. To facilitate efficient use, the database supports a range of search and filter options aligned with fact-checkers’ preferences.

6.2 Video annotation interface

When fact-checkers choose to annotate a specific video in the annotation database, clicking the corresponding ‘annotate’ button directs them to a dedicated annotation interface (Figure 8). The interface includes four key components. First, it features an embedded YouTube player on the left, allowing seamless access to the video during annotation. Second, a video profile section with textual metadata like transcript, subtitles, title, and description in separate tabs. These metadata elements are selected in consultation with fact-checkers for their importance in veracity analysis. Third, the annotation interface enables fact-checkers to curate information and record investigative notes. Annotations are added by selecting text in the metadata tabs and clicking the ‘annotate’ button. This action highlights the selected text and opens an annotation box with fields for entering comments, sources, a veracity label for the claim, and descriptive tags (keywords describing the claim). Tags aid in categorization, organization, and enhancing the searchability of claims. The fourth component is the overall veracity label for the entire video. Notably, individual claims and the overall video may have distinct veracity assessments. The veracity labels used by Pesacheck include categories such as altered, false headline, hoax, missing context, partly false, satire, false, likely false, and no conclusion. The ‘No conclusion’ label is also used as a default when a video has not yet been formally fact-checked. Determinations about whether to fact-check a video, which claims to prioritize, and what labels to assign are left to the discretion of expert fact-checkers.

6.3 Claims database

Fact-checkers traditionally manage spreadsheets to document their fact-checking work, but expressed a desire for a more automated and structured approach. YouCred’s claim database (Appendix Figure A.8) addresses this need by serving as a centralized repository of misinformative claims identified within annotated videos. Each entry includes the claim text, its veracity label, associated tags, the name of the fact-checker, and the date of annotation. Beyond streamlining documentation, the database enables fact-checkers to identify recurring narratives and patterns, supporting longer-term educational and policy-oriented responses.

7 Evaluate stakeholders’ acceptance

We deployed YouCred at Pesacheck for nine months, beginning in September 2022. To ensure effective adoption, we conducted two organization-wide training sessions on November 7, 2022, and March 13, 2023, and provided a demo video and comprehensive documentation. During the deployment period, the research team provided technical support and iterative improvements, with

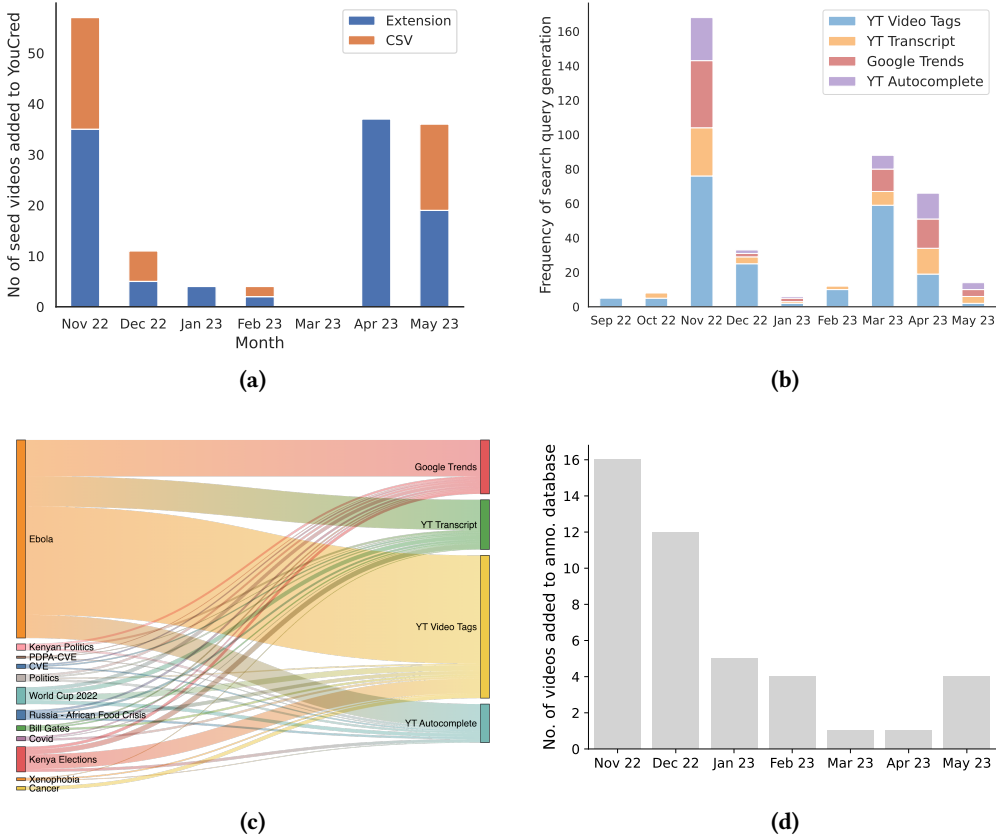


Fig. 9. Figure (a) illustrates the number of seed videos added to YouCred through manual CSV uploads and the use of the ‘YouTube-CSV-Helper’ extension. Figure (b) presents the usage frequency of YouCred for generating search queries using the four proposed methods throughout the 9-month deployment period. Figure (c) provides an overview of the topics monitored using YouCred and the corresponding proportions of query generation methods utilized for each topic. Figure (d) illustrates the number of potentially misinformative videos added by fact-checkers to YouCred’s annotation database.

the system and its documentation transitioned to the organization at the end of the study. We evaluated YouCred’s acceptance throughout the deployment period by analyzing its usage patterns and conducting interviews with the fact-checking community. We discuss the results below.

7.1 Patterns of Usage over Time

7.1.1 Uploading seed videos for topics. Figure 9a presents the number of seed videos added via CSV and YCH extension during the deployment period⁷. In the initial month of YouCred’s deployment, system usage was limited. Feedback from fact-checkers indicated that manual CSV creation was a barrier to usage. To address this, we developed and launched the YCH extension in late October 2022. The extension significantly improved usability and adoption—fact-checkers used it far more

⁷We began logging the usage of the extension and CSV uploads in November, coinciding with the organization’s adoption of the extension.

frequently than the manual CSV option. In total, 124 unique videos were uploaded, out of which only 73 were present on YouTube at the time of writing this paper.

7.1.2 Search query generation. Figure 9b shows how often YouCred was used to generate search queries during the 9-month deployment period. Usage spiked in November following the extension's release. The initial spike is consistent with exploratory use following the release of a new tool, as fact-checkers tested YouCred's capabilities. Usage declined in January and February, likely due to two external disruptions: Pesacheck's holiday office closure (Dec 15–Jan 10) and a server outage in February. We observed an increase in usage following a second organization-wide training session conducted in March. While activity remained lower than the initial November peak in subsequent months, fact-checkers continued to use YouCred across a range of topics and query generation methods. Usage patterns over time also surfaced evolving needs and areas for improvement, as well as constraints introduced by external platform dependencies (e.g., YouTube API quota restrictions discussed in Section 7.2.3). In addition, support for the system concluded in March as the project was handed over to the organization, which may have further influenced usage levels. Together, these dynamics highlight the practical challenges of sustaining usage, maintenance, and continuous improvement in real-world newsroom deployments, which we discuss further in Section 8.5.

Overall, we observe that among the four query generation methods, video tags were used the most, followed by Google Trends and the transcript method. Note that YouCred does not employ default methods. In each usage, fact-checkers must select the methods they wish to utilize explicitly.

7.1.3 Topics monitored. Figure 9c presents the range of topics monitored on YouCred and the proportion of query generation methods used for each topic. We observed that the topic of Ebola was the most monitored, followed by the Kenyan elections and the FIFA World Cup. In the later stages of deployment, fact-checkers expanded their focus to include newer topics such as the Personal Data Protection Act (PDPA)⁸⁹ and Countering Violent Extremism (CVE) policies¹⁰.

7.1.4 Video Annotation and Credibility Assessments. Figure 9d demonstrates the number of videos added to YouCred's annotation database during the deployment period. In total, 45 videos were assessed for credibility. Please note that before YouCred, fact-checkers at Pesacheck rarely searched for misinformative content directly on YouTube. Instead, videos were typically encountered when embedded in posts on platforms like Facebook or X. YouCred enabled more direct, systematic engagement with YouTube content for the first time, and the 45 annotations represent a shift from incidental to intentional monitoring of the platform.

7.1.5 Similarity of misinformative videos discovered via YouCred with the seed videos. We analyze whether the videos fact-checkers annotated through YouCred introduced new misinformative claims or echoed those found in the original seed videos. At the time of analysis, 16% of the seed videos and 9% of the annotated videos were no longer available on YouTube. For the remaining videos, we combined available descriptions and transcripts and used GPT-4¹¹ to extract key claims from each. The model was prompted to identify prominent assertions within each video's content. The first author then qualitatively compared claims across the two sets to assess thematic overlap. Only 9.75% of annotated videos contained claims similar to those in the seed videos, suggesting that fact-checkers primarily used YouCred to surface new misinformative claims, rather than focusing solely on videos similar to the original seed videos.

⁸<https://www.dataguidance.com/notes/south-africa-data-protection-overview>

⁹<https://www.dataguidance.com/jurisdiction/africa>

¹⁰<https://www.usaid.gov/policy/countering-violent-extremism>

¹¹<https://openai.com/index/gpt-4/>

7.1.6 Limitations. The quantitative analysis conducted to assess YouCred faces several limitations. Prior to YouCred, Pesacheck’s fact-checkers rarely searched for misinformation directly on YouTube; instead, videos were typically encountered through social media posts or user-submitted tip lines. As a result, we lack baseline data on the time and effort previously spent forming search queries, limiting our ability to make direct pre-/and post-comparisons on efficiency. We also refrain from comparing the videos discovered for fact-checking before and after the deployment. This is partly because YouCred was not designed to replace existing methods of video discovery but to complement them by enabling more intentional and structured discovery on YouTube—an area fact-checkers identified as a blind spot. Furthermore, the validity of a direct comparison is compromised by various factors, such as shifting editorial priorities and significant real-world events. Despite these constraints, YouCred’s sustained usage over nine months, along with qualitative feedback from the fact-checking team, provides evidence of its practical value and relevance in Pesacheck’s fact-checking workflows.

7.2 Semi-structured interviews

To gain a comprehensive understanding of YouCred’s adaptation and usage, we complemented the quantitative evaluation with semi-structured interviews. We provide the details of the evaluation interviews below.

7.2.1 Participant sampling technique. We utilized a purposive sampling technique [66] to select participants for the semi-structured evaluation interviews. We reached out to fact-checkers who regularly used YouCred, identified from the video annotation database and extension log. Additionally, we included other stakeholders who were actively involved in the design process, even if they did not extensively use the tool. Such individuals included managers and fact-checking editors. Our sample consisted of eight individuals from Pesacheck (referred to as I1-I8).

7.2.2 Interview Protocol and Data Analysis. We began the interviews by requesting participants to provide a brief overview of their previous methods for monitoring the YouTube platform before the deployment of YouCred. This initial step allowed them to delve into the constraints they faced with their existing methods, laying the groundwork for discussing how YouCred addressed these limitations. Subsequently, we delved into the integration of YouCred into their fact-checking workflow, exploring specific scenarios and topics where they utilized the tool. Participants were then asked to demonstrate how they monitored a particular topic using YouCred, guiding us through their step-by-step process and emphasizing the distinctions between direct searches on YouTube and the use of YouCred. To facilitate direct comparisons, we also displayed YouTube and YouCred interfaces side by side on the screen, prompting participants to share insights into the differences and how these disparities influenced their fact-checking process. Throughout this interactive process, we delved into YouCred’s effectiveness and discussed how the system aligns with participants’ specific needs. We also actively encouraged the interviewees to provide suggestions for improving YouCred and discuss any limitations they may have encountered during their usage. To ensure comprehensive and unbiased feedback, participants were consistently prompted to provide detailed responses and to cite specific examples [21]. We also implemented an anonymous feedback form within the YouCred system to gather additional feedback about the system and took precautions to avoid leading questions during the inquiries.

The semi-structured interviews, lasting approximately 60 minutes each, were conducted remotely over Zoom by the first author, with participants’ consent. Throughout the interviews, the first author made detailed notes based on observations. Subsequently, the first author transcribed all the interviews and notes and thematically analyzed them using an inductive coding scheme [12].

7.2.3 Findings. In this section, we present the themes emerging from the final interviews.

Fact-checkers have integrated YouCred into their day-to-day workflow: Fact-checkers revealed that YouCred “comes in handy because of the lack of available free tools to look for misinformation online, especially on Youtube” (I3). The majority of participants revealed that as part of their routine work, whenever they find misinformative videos, “on Facebook or Twitter”, they “try to feed it into the YouCred system” (I8) to discover additional misinformative videos about the topic.

YouCred provides rich and meaningful query suggestions: YouCred’s query generation methods proved to be a valuable resource for fact-checkers providing “a pool of information that would otherwise not be available if we [they] were generating queries manually” (I4). Fact-checkers highlighted the effectiveness of YouCred’s integration with Google Trends which consistently delivered “high-quality search results for them” (I3). Furthermore, the use of video tags exposed them to conspiracy terms associated with topics they might not have discovered otherwise. As I7 noted, “I probably would not know of these things anyone would relate them to Ebola..I would have searched Ebola with keywords health, Africa, pandemic, and not necessarily New World order, or Illuminati”. Additionally, video tags unveiled strategies employed by content creators to boost video popularity, such as “using the names of celebrities as tags so that the video can trend” (I4).

Fact-checkers also appreciated the flexibility to customize their searches on YouCred. As I3 expressed, “I like YouCred suggesting keywords and also being able to edit those keywords and also adding your own, because that refines your search a lot and leads to a higher possibility of getting what you want with the YouCred tool but with the YouTube keyword search, it’s a hit or miss”.

YouCred streamlines credibility assessment process: Participants revealed that the annotation interface simplified and streamlined their workflow. I7 explains—“I really like how you can highlight claims, you’re able to tag what category the video falls in.. and also add evidence all in one place..I don’t have to document it on Google docs..fill in the columns...you can do all of that within the tool itself..its really important and just shortens the entire process” (I7). I1 also finds the annotation interface valuable since it empowers them to dissect videos into individual claims, facilitating focused and thorough investigations.

“When we look at multiple items within the video that might be false, YouCred allows us to break down the video into multiple claims and investigate each section” - I1

Fact-checkers use YouCred’s claim database as an informative resource: Fact checkers’ consider YouCred’s claim database as a powerful resource that goes beyond simply cataloging misinformative claims. For example, I4 suggested that the tool helps them avoid duplication of effort since “you can realize that you know, someone else has had really added the video [in the claim’s database]..and is working on it”. In addition, they find the addition of channel details on the page useful as “it reveals the frequent spreaders of misinformation...So you can actually go to their channel and see, if have they shared any more misinformation” (I8). I2 also shared how tags in the claim database allow them to “quickly identify clusters of videos discussing the same claim or topic”, helping them determine its popularity efficiently. In the future, I4 suggested integrating the claims database directly into the YouTube platform, by showing disclaimers or fact-checks with the timestamps associated with the misinformative claims as potential intervention on the platform.

Youcred enhances the efficiency of fact-checking workflow: Participants revealed that using “YouCred tool, [they] could spend less time getting misinformation as opposed to when they were searching

on YouTube directly” (I5). They also talked about several features that have played a vital role in improving the efficiency of their fact-checking workflow. According to I2, the preview mode on the *view-results* page is a valuable tool that allows fact-checkers to do “*quickly preview [the video] on the same page and decide if they even want to annotate a video in the first place without going to Youtube.*”. I3 highlights the usefulness of the ability to track multiple search queries on the *view-results* page by stating, “*we are now have everything on one page, and we’re able to monitor everything at the same time...instead of doing multiple searches in several tabs..it saves you more time*”. Additionally, the interactive graphs also garnered positive feedback. Participants expressed that “*working with graphs is easier as opposed to filtering search results manually as it gives a visual touch which is preferred*” (I8).

Limitations and opportunities: Throughout the interviews, participants shared insights into challenges encountered while using YouCred, offering valuable suggestions for its further enhancement. A primary challenge highlighted by most participants was the restrictive YouTube API quota limit, significantly impeding their workflow. I4 elaborates—“*The problem I have been facing with the API key is that it’s not able to sustain in some cases, even a single query. That means now, if it says that the API key is not able to do something, then you have to wait until 24 hours have lapsed before you start using it again.*” Currently, YouCred relies on the limited YouTube API quota (10,000 requests/day), which often gets exhausted quickly. To address this, we have applied to join the YouTube Researcher Program¹², which offers expanded access to the YouTube API, though our application has not been approved to date.

Participants conveyed that they find the process of curating seed videos laborious. They highlighted the practice of incorporating videos discovered through YouCred into the seed set to broaden the seed numbers. Despite this, they expressed a desire to “*figure out a way where [they] don’t have to feed YouCred with content for it to give it back related misinformation*” (I8). Participants suggested several ways to reduce the labor of inputting videos. One suggestion was integrating YouCred with user tiplines, where user-reported videos could both surface emerging topics and serve as alternative seed videos. We also envision extending YouCred’s deployment to other fact-checking organizations, fostering a network for sharing resources (such as user-reported YouTube videos) and facilitating knowledge exchange. In addition, participants recommended expanding YouCred’s functionality to include the capability to identify YouTube videos that are “*trending or widely shared across other platforms*” (I3). This enhancement would allow YouCred to incorporate these popular videos as seed content automatically. Overall, these potential seed video sourcing strategies illustrate how YouCred could leverage a broader assemblage of existing tools, platforms, and practices.

Participants further expressed a need to “*refine search results.. and filter out the noise [non-misinformative videos] in the YouTube search results*” (I3). They pointed out that YouTube’s top search results predominantly come from authoritative sources, which are less likely to contain misinformation. Consequently, they find themselves sifting through many videos to identify user-generated misinformative content. In our future iterations of YouCred, we plan to refine search results by employing machine learning algorithms to identify and prioritize results with a higher probability of containing misinformation. One potential way is to look for linguistic signals in the videos’ comments to identify potentially misleading videos [54]. Finally, participants also suggested design changes in YouCred. I4 recommended showing all the suggested queries in the drop-down menu on the *view-results* page so that they don’t have to go back to the methods page in case they want to select new keywords. Additionally, I2 suggested incorporating a tooltip with the search query text box, featuring all possible search operators supported by YouCred for user convenience.

¹²<https://research.youtube/>

These suggestions underscore the need for active maintenance and evolution in real-world system deployments to align with the evolving requirements of stakeholders.

8 Discussion

In this section, we reflect on the design implications of our work and discuss how we need to think about the maintenance of socio-technical systems beyond the initial deployment period.

8.1 Design and policy implications for video platforms

Monitoring video search platforms such as YouTube presents distinct challenges compared to traditional social media platforms like Facebook or Twitter (now X) [59]. Unlike these platforms, YouTube offers limited native affordances, such as trending topics, topic-level trending videos, multi-query tracking, or region-specific filters, that could otherwise support systematic identification of misleading content. As a result, fact-checkers continue to rely on labor-intensive manual searches, crafting queries based on guesswork and domain knowledge. Our work reveals that these limitations are not merely usability gaps but reflect broader design shortcomings in how video platforms support fact-checking work. Features that fact-checkers consistently valued in YouCred, such as tracking multiple search queries in parallel, filtering results by region and time, and monitoring engagement trajectories could be natively supported by platforms like YouTube. Comparable precedents already exist, for example, TweetDeck¹³, an application owned by X, consolidates information into a single dashboard through customizable columns. It allows fact-checkers to track multiple feeds, hashtags, or accounts in parallel, and apply filters for real-time monitoring. This tool has gained widespread adoption among fact-checkers globally [71]. A similar interface tailored for video fact-checking would significantly enhance the capacity of fact-checkers worldwide.

Our deployment also illustrates how platform-controlled APIs can shape the feasibility and reliability of real-world fact-checking work, particularly in the absence of official tooling or sustained platform support. YouCred deployment depended on YouTube APIs, which are subjected to restrictive rate limits. When quotas were exhausted, fact-checkers were forced to pause work for up to 24 hours, an especially consequential disruption during time-sensitive events such as elections or public health crises. These constraints are not unique to YouTube. Across platforms, APIs are often restrictive, expensive, or completely unavailable, shaping what forms of misinformation research is practically possible [22, 60]. This fragility is further amplified by broader shifts in the platform ecosystem, including the rollback of formal fact-checking programs [81], and the withdrawal of funding from this space [29]. Together, restrictive APIs and unstable institutional support create precarious conditions for sustained misinformation monitoring, highlighting the need for platform-level policies and researcher access programs that recognize fact-checking as critical public-interest infrastructure rather than a peripheral use case.

8.2 Design implications for fact-checking tools and infrastructures

Our work addressed a key gap in video fact-checking by providing a structured approach to search query generation. Specifically, YouCred used seed video content as a catalyst for generating new queries, turning small curated sets of videos into scaffolds for broader monitoring. This approach has broader relevance beyond YouTube, as the resulting queries can be reused to track similar narratives and claims across platforms. However, our findings also surface an important tension in this seed-based design approach. While seed-based query generation was useful in practice, fact-checkers found the task of curating seed videos taxing. Yet the effectiveness of the system depended on having high-quality seed content, revealing a trade-off between structured discovery and the

¹³<https://en.wikipedia.org/wiki/Tweetdeck>

labor required to initiate it. This tension points to a key design implication for misinformation tools that rely on human-curated inputs: systems must explicitly account for the real costs of seeding and bootstrapping, particularly for emergent topics where few known examples exist. One way to reduce this burden is to draw seed content from existing organizational workflows, such as user tiplines where videos are reported by audiences, or content encountered through cross-platform monitoring, rather than requiring all seeds to be manually curated within the system itself.

YouCred also presents an opportunity to develop a shared resource across fact-checking organizations. Multiple groups deploying the system could collectively build repositories of seed videos, claims, annotations, and investigative resources. Such repositories would not only reduce duplicated effort but also provide valuable data for studying the circulation of misinformation in the Global South, where empirical research remains limited. At the same time, shared infrastructures introduce non-trivial challenges. Collaborative access raises questions around governance, trust, access control, and the risk of data pollution, sabotage, or delegitimization in highly contested information environments, underscoring the need for careful institutional design. Any move toward shared repositories would therefore require carefully designed participation boundaries, such as restricting access to established networks like IFCN-affiliated organizations, and explicit institutional oversight, rather than open collaboration by default.

Finally, YouCred's design points to broader implications for how fact-checking tools could connect to downstream interventions. For example, YouCred's annotation database could connect seamlessly with Pesacheck's Twitter Debunk bot¹⁴, which replies to tweets containing misleading URLs with corrective fact-checks. By linking the two systems, the bot could recognize when a tweet shares a video already annotated in YouCred as containing misinformation and automatically post the corresponding fact-check. Such intervention support is not limited to Pesacheck. A browser extension, for instance, could display warnings when users encounter misinformative videos that have already been fact-checked. Drawing inspiration from Hughes et al.'s design for embedding citation signals [47], YouCred's claims database could be leveraged to surface fact-checks directly within the video timeline. For example, displaying corrective information in a pop-up at the timestamp when a false claim is discussed, much like how YouTube overlays advertisements.

8.3 Balancing automation and human expertise

Our work also adds to the growing body of research on human-machine collaboration. Prior studies have examined the complexities of human-computer integration, asking which tasks are best automated, how users perceive the role of automation, and how the division of labor between humans and systems can be negotiated [6, 28, 68]. A consistent theme across this literature is the inadequacy of fully automated approaches as the ultimate solution in high-stakes domains. In the context of misinformation, fact-checkers themselves have expressed skepticism about fully automating the process of determining veracity, emphasizing that a human presence is essential for accurate and effective fact-checking [59, 71]. Our findings align with this perspective and highlight the nuanced requirements of fact-checkers. Pesacheck's fact-checkers insisted on retaining agency over critical decisions such as selecting videos for review and assigning veracity labels, but they also desired automated systems that could support their efforts by suggesting potentially misleading content. YouCred was designed with this balance in mind: it scaffolds fact-checkers' work by generating candidate queries while preserving their autonomy in deciding what to search and which videos to investigate. By foregrounding autonomy while offering targeted support, YouCred

¹⁴<https://twitter.com/debunkbotafrica?lang=en>. Pesacheck created a Twitter bot named debunk bot that detects tweets containing URL(s) to misinformative content and replies to them with the fact-check debunking the false claim.

illustrates a design stance that balances automation with agency, advancing ongoing conversations about the future of human–machine collaboration in fact-checking.

8.4 Bridging design-reality gap in design of fact-checking systems

Prior work in human–computer interaction (HCI) and computer-supported cooperative work (CSCW) has long emphasized participatory design and community engagement as foundations for building sustainable socio-technical systems [15, 56, 59]. In this work, we adopted participatory methods to design a fact-checking system for YouTube in collaboration with stakeholders at Pesacheck across the full design–build–deploy cycle. Over the course of two years (June 2021–May 2023), this collaboration was sustained through regular meetings, interviews, and discussions with fact-checkers. These engagements fostered meaningful dialogue, surfaced situated knowledge, and deepened our understanding of fact-checkers’ day-to-day concerns and practices, illustrating the well-established benefits of participatory approaches. However, sustaining this collaboration required continuous relational work. Building and maintaining trust, aligning expectations, and nurturing ongoing communication demanded a considerable investment of time and effort. This emphasis on sustained trust-building and listening resonates with a broader body of work in HCI that frames community relationships as an ongoing commitment rather than a one-time design activity [20, 37, 67]. Justice-oriented and community-centered design scholarship has argued that meaningful participation depends on long-term engagement, attentiveness to local values, and respect for situated expertise [20]. These works have also highlighted that technologies are inseparable from the social and cultural contexts in which they are deployed, and that maintaining community authority over interpretation and use is central to ethical system design [37, 67]. In this sense, the relational labor involved in our collaboration was not ancillary to system development, but foundational to bridging the gap between design intentions and real-world practice. Yet despite its centrality, such relational work often remains invisible in accounts of socio-technical systems, rarely acknowledged as part of design or research, and is usually not budgetable in grant proposals [63]. However, this labor is central to system adoption and longevity [15, 62]. Recognizing and supporting these efforts is essential for ensuring the long-term sustainability of collaborative initiatives and enhancing their broader impact. YouCred’s development was not solely driven by the aim of meeting stakeholders’ needs; it went a step further by building upon the assets of fact-checkers. Fact-checking stakeholders weren’t passive users; their expertise and experiences directly shaped the system’s functionalities and features. Asset-based approaches foreground what participants already do well and what infrastructures they already maintain [39, 92, 101, 103]. The implication of this asset-oriented perspective is twofold. First, it enables systems to integrate more seamlessly into organizational routines, reducing the friction that often undermines technology adoption. Second, it shifts the politics of design: rather than positioning fact-checkers as beneficiaries of external expertise, it recognizes them as knowledge-holders and infrastructural contributors in their own right. Our case demonstrates how fact-checkers’ assets—ranging from multilingual expertise, investigative heuristics annotation practices—can be mobilized as building blocks for new tools, pointing toward a broader agenda of designing fact-checking infrastructures that strengthen rather than overwrite local capacities.

8.5 Maintainability of socio-technical systems

Deploying a system in the real world is a complex undertaking that goes beyond the initial development phase. It requires ongoing maintenance and continuous improvement to ensure its effectiveness and relevance in the ever-evolving technological and information landscape [90]. As we deployed YouCred, we also encountered various challenges, including scalability, concurrency, and the need for real-time bug handling to ensure uninterrupted system usage. These experiences

reinforced a broader insight: building and deploying a system is not sufficient on its own. Deployed systems must continue to adapt as user needs evolve, new requirements emerge, and system limitations become visible in practice. Sustained use, therefore, depends on active maintenance, iterative improvement, and long-term support. This underscores the significance of investing resources and efforts into the ongoing maintenance of deployed systems.

In recent literature, scholars have raised concerns about sustaining maintainability efforts after the initial project funding ends [63, 90]. Maintenance and repair activities have been recognized as overlooked yet essential aspects of socio-technical initiatives, encompassing creativity, innovation, knowledge, power dynamics, and ethics of care [51]. Efforts have been made to study the challenges of sustaining and maintaining the outcomes of Human-Computer Interaction (HCI) design and systems projects beyond their runtime and beyond the researchers' role within the project context [23, 51, 52, 63]. For instance, Meurer et al. suggest the importance of nurturing a sense of ownership among research participants from the project's outset and developing their technological capabilities [94]. However, the feasibility of such approaches varies greatly depending on contextual factors, including the technical and human resources available to different researchers and stakeholders for maintaining the designed technological artifacts. Thus, we believe that active support from funding bodies is crucial to encourage and enable researchers by allocating time and resources specifically for maintenance activities, ensuring the continuity of these efforts beyond the project's duration.

9 Conclusion

In this paper, we build YouCred, a fact-checking system to help fact-checkers with misinformation discovery and credibility assessments on YouTube. The system is a result of a 2-year collaboration with the Pesacheck fact-checking organization. To design YouCred, we adopted participatory methods where the goal was to build a system that not only caters to stakeholders' needs but also utilizes the assets of the community. YouCred helps fact-checkers find misinformation by suggesting search queries related to misinformative topics and provides an interface to enhance the efficiency of credibility annotations. To evaluate YouCred's effectiveness, we deployed the system at Pesacheck, monitored its usage for nine months, and conducted follow-up semi-structured interviews. Evaluations revealed that YouCred was positively received by the fact-checking community who acknowledged the system's utility and effectiveness.

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A Appendix

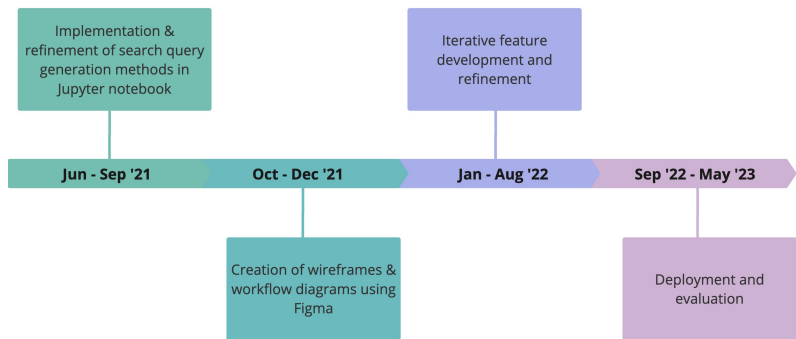


Fig. A.1. Timeline of YouCred’s design, development, and deployment process

Home Get Started Topic Database Video Annotation Database Blocklist Database Site Analytics Feedback

Topic Database

Download Topic Data Refresh Data

Added Time	Topic	Video ID	Video Title	Misinformation Category	Annotate	Remove
Mon, 21 Nov 2022 22:32:22 GMT	bill gates	u4o6L80v28w	Bill Gates Tells World Leaders 'Death Panels' Will Soon Be Required	False	Annotate	✖
Wed, 23 Nov 2022 01:16:46 GMT	cancer	gWQPYfbc0dw	() CURE FOR ALL TYPES OF CANCER, THE SECRET IS IN HOT WATER WITH SLICES OF PINEAPPLE	False	Annotate	✖
Fri, 18 Nov 2022 16:01:09 GMT	cancer	WcF9PCkvd0U	Finally We Have A Cure For Cancer - Japanese Scientist	Partly False	Annotate	✖
Fri, 12 May 2023 06:48:16 GMT	CVE	u48CExh8H	ظهور الحركة آزاد الإسلامية الحركة الأولى لجمدة آزاد	No Conclusion	Annotate	✖
Fri, 12 May 2023 06:48:16 GMT	CVE	kchzn3gn0MQ	قوات الحركة الوطنية للحرير آزاد في منظر جليل يرافق في إطار عملية التسجيل العسكري	No Conclusion	Annotate	✖
Mon, 15 May 2023 01:32:33 GMT	CVE	pKXvaWT9gmM	آزاد : إحتفال الشبية حوسبة لإختار الإستراتيجي القائم لقائين السكان ويستكثهم	No Conclusion	Annotate	✖
Fri, 12 May 2023 06:48:16 GMT	CVE	BgcWwh0eQ	كانت جماعات مسلحة تأسسها بالسلطان على التماسح في جماعة واحدة تقرير محمد حاميدو كاتيتي	No Conclusion	Annotate	✖
Fri, 12 May 2023 06:48:16 GMT	CVE	5qbyZeh3hUA	قصة بداية القردة آزاد	No Conclusion	Annotate	✖
Mon, 15 May 2023 01:32:33 GMT	CVE	t6U7glud-c0	عالم جماعة نصرة الإسلام والمسلمين في كلمة كليب العسكرية	No Conclusion	Annotate	✖
Fri, 12 May 2023 06:32:24 GMT	CVE	tUwfw8b0C8X	في مدينتها أثناء الاحتفال بالذكرى العاديا على إعلان إستقلال دولة آزاد MNEA جزء من كلمة منسق مكاتب	No Conclusion	Annotate	✖

Fig. A.2. Snapshot of YouCred’s topic database that displays all the seed videos uploaded by fact-checkers along with their corresponding topics. The videos can be easily deleted from the database or sent to the annotation database for credibility assessments.

Generated YouTube Search Query*
You may edit the generated search query as you wish.

Ebola AND ("black genocide" OR "new world order")

(Optional) Type to select a region and desired search result language on YouTube

☒ Worldwide

Type to select a region

- ☐ AF Afghanistan
- ☐ AX Aland Islands
- ☐ AL Albania
- ☐ DZ Algeria
- ☐ AS American Samoa
- ☐ AD Andorra
- ☐ AO Angola
- ☐ AI Anguilla
- ☐ AQ Antarctica
- ☐ AG Antigua And Barbuda

All Languages (You would be able to choose three languages at the same time)

Type to select a result language

- ☐ aa Afar
- ☐ ab Abkhazian
- ☐ ae Avestan
- ☐ af Afrikaans
- ☐ ak Akan
- ☐ am Amharic
- ☐ an Aragonese
- ☐ ar Arabic
- ☐ as Assamese
- ☐ av Avaric

(Optional) Select video publish date range*

From 01/01/1970 to 08/31/2023

Number of Desired Video Results (1-200):*

25

Replace blocked video:

☐

Order of Results:*

☒ Uploaded Date ☐ Video Rating ☐ Relevance ☐ Number of Views

Next

Fig. A.3. YouCred enables users to filter search results by location, country, and publication date. Additionally, it provides the option to sort search results based on metrics such as relevance, video count, etc.

Select one or more frequently occurring keywords extracted from the provided videos to fetch Google Trends topics*

Select top % of keywords: 5

Sort tags *

kenya elections ruto news raila secret uhuru chebukati

Select bottom % of keywords: 2

Sort tags *

wib offer wib raila williamruto williamruto ralaodinga women

(Optional) Add additional custom keywords (separated by comma)

Save

Select one or more Google Trend topics

The tool finds the most popular search queries related to topics you select below.

Please note: Google Trend may not have any autocomplete suggestion for your keyword.

Wafula Chebukati William Ruto Elections in Kenya Kenya Country in East Africa

(a)

(Optional) Select one or more countries and languages of interest

Popular search queries vary between countries and languages

☒ Countries monitored by Pesachick

☒ Worldwide

Search for a country (optional)

- ☐ DZ Algeria
- ☐ AO Angola
- ☐ BJ Benin

All Languages

Search for a language (optional)

- ☐ aa Afar
- ☐ ab Abkhazian
- ☐ ae Avestan

(Optional) Select date range of interest

Since popular search queries vary across time, please select the date range you're interested in.

From 06/01/2022 to 06/12/2023

Generate

Here are some top and rising search queries on Google Trend

You may select one or more queries to generate a YouTube search query

Top Search Queries Sort tags (by Frequency) *

president william ruto president ruto raila odinga ruto today william ruto today raila odinga uhuru

Rising Search Queries Sort tags (by Frequency) *

chekbukati death cherera chebukati family where is chebukati now wafula chebukati wife raila odinga

Generated YouTube Search Query*

You may edit the generated search query as you wish.

kenya elections AND ("wafula chebukati" OR "raila odinga" OR "is chebukati alive" OR "chekbukati death" OR "president william ruto")

(b)

Fig. A.4. Figures (a) and (b) depict YouCred's query generation page utilizing the Google Trends (GT) method. (A) Fact-checkers begin by selecting keywords that serve as seed words for extracting GT topics. The seed words are the most and least occurring unigrams and bigrams found in the titles and descriptions of seed videos related to the topic. (B) They also have the flexibility to add custom keywords. On entering seed words in the search bar of GT, the platform presents a dropdown list of existing GT topics containing that word. (C) All such topics are presented to users, who can select the topics of interest. (D) Next, fact-checkers select the countries and languages they want to focus on and (E) specify the desired date range. (F) YouCred then extracts the GT search queries using the specified parameters, which fact-checkers can review and select from. (G) All search queries selected by fact-checkers are appended by the OR (+) operator to ensure that the search results obtained encompass the chosen queries.

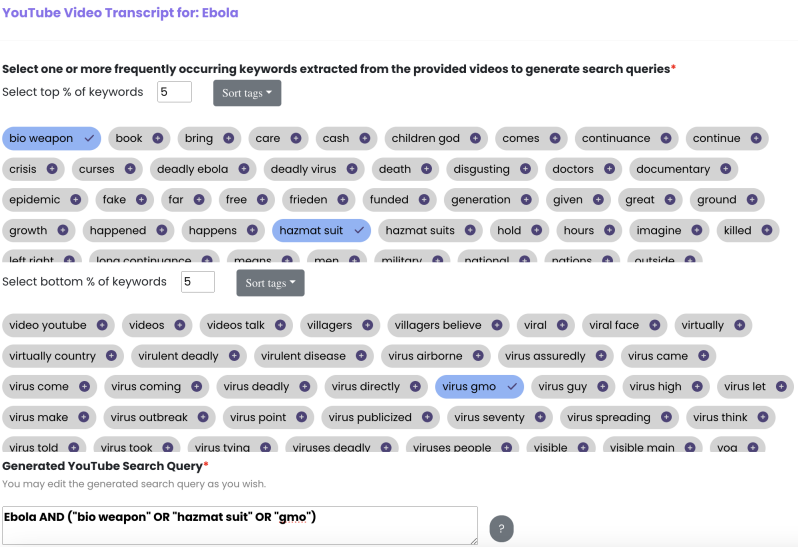


Fig. A.5. Figure illustrating YouCred’s query generation method page utilizing the transcript method. The interface suggests x% of most and least occurring words and phrases in transcripts to be utilized as potential search queries.

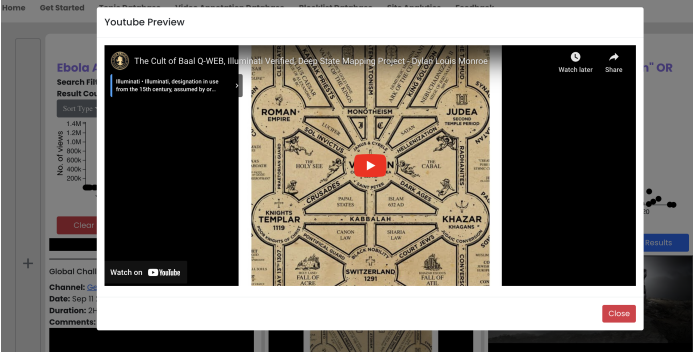


Fig. A.6. Snapshot of YouCred’s preview mode. Fact-checkers can click on any video in the *view-results* page and can view the video in the system itself.

VideosClaims

Download Videos Table Data

Refresh Data

Filter by Fact-checker: -----All-----

Filter by Conclusion: -----All-----

Filter by Topic: -----All-----

Select the range of Upload Date

Select the range of Added Date

From 01/01/2005to 06/06/2023

From 01/01/2022to 06/06/2023

Video Title

Search Video Title

Annotate	Video title	Fact-checker	Conclusion	Views	Likes	Upload date	Channel	Topic	Tags	Added date
Annotate	KALONZO FINALLY JOINS RUTO #ruto #azimio aumojala #kenyakeganga #supremecourt #uhuru #ralia #martha		False Headline	9.1K	39	9/7/2022	omdigizi	kenya elections	kenyan election	11/17/2022
Annotate	BREAKING NEWS!!!WAJACKOYA AND KALONZO JOINS UDA PARTY?PRESIDENT RUTO RECEIVES THEM SERIKAU IMEJAA?		False Headline	1.2K	12	9/4/2022	KK Ngare 254	kenyan politics		11/29/2022
Annotate	UK PM Rishi Sunak performing religious rituals before entering his new office 10 Downing Street...		Missing Context	854	4	10/25/2022	POLITICAL BANTER	politics		11/29/2022
Annotate	Children Reciting The Holy Quran On The Opening Ceremony Of FIFA World Cup 2022, Qatar #FIFAWorldCup		False	29.6K	1.0K	11/19/2022	Bannuzzan Production	world cup 2022		11/29/2022
Annotate	Αυτή είναι μια άγρια δόση το βόλεϊς Κερασε (Dough) που δίνει εκδήλεις του ορέ		False	14	2	5/18/2023	elena111fuldeliverusfromevil	terrorism		5/21/2023

Fig. A.7. Figure shows the snapshot of YouCred’s video annotation database. Fact-checkers add videos to this database while exploring the *view-results* page or directly from the browser extension. All videos have a corresponding annotate button, which takes the fact-checkers to the video’s annotation interface. This database contains the video’s title and other metadata. The conclusion column is populated once fact-checkers assign a veracity label to the video on the annotation interface. The added date column denotes the date the video was added to the database. The page also provides a variety of search and filter options to find or view selected videos.

VideosClaims

Download Claims Table Data

Filter by Fact-checker: -----All-----

Filter by Conclusion: -----All-----

Filter by Tags: -----All-----

Select the range of Added Date

From 01/01/2022to 06/06/2023

Video Title

Search Video Title

Video title	Fact Checker	Time stamp	Claim	Conclusion	Tags	Added Date
BREAKING NEWS!!!WAJACKOYA AND KALONZO JOINS UDA PARTY?PRESIDENT RUTO RECEIVES THEM SERIKAU IMEJAA?			BREAKING NEWS!!!WAJACKOYA AND KALONZO JOINS UDA PARTY?PRESIDENT RUTO RECEIVES THEM SERIKAU IMEJAA?	False Headline	George Wajackoyah, Kalonzo Musyoka, UDA Party	12/6/2022
UK PM Rishi Sunak performing religious rituals before entering his new office 10 Downing Street...			UK PM Rishi Sunak performing religious rituals before entering his new office 10 Downing Street...	Missing Context		12/1/2022
Children Reciting The Holy Quran On The Opening Ceremony Of FIFA World Cup 2022, Qatar #FIFAWorldCup			Opening Ceremony Of FIFA World Cup 2022, Qatar	False Headline	2022 World Cup Opening Ceremony, Quran recitation	11/29/2022
TUTASHIKA RAILA TUFUNGIE KAMITI NDO ASHIKE ADABU NA AWACHE KUTUSI RUTO-GACHAGUA.			TUTASHIKA RAILA TUFUNGIE KAMITI NDO ASHIKE ADABU NA AWACHE KUTUSI RUTO-GACHAGUA.	False Headline		12/2/2022

Fig. A.8. Snapshot of YouCred’s claim database that stores entries for all the misinformative claims highlighted by fact-checkers in the annotated videos along with the video title, veracity label, tags associated with the claim, and the date when the video was added to the annotation database.

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